Worksheet 8

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10/06/2026

1 W 8-1

The initial cashflow is \$100000, the project's life is 5 years, we expect $100,000$ cupcakes to be sold per year with an expected price of \$0.75 per unit sold and an expected variable cost of \$0.25 per unit sold. Annual fixed costs are $\$10,000$ and taxes are 30%.

Year	0	1-5
Sales		75000
- Costs		−35000
- Depreciation		−20000
EBIT		20000
- taxes		−6000
Net Income		14000
+ Depreciation		20000
Operating CF		34000
- capital spending	−100000	0
- change in NWC		0
Project CF	−100000	34000

NPV:

$$\mathbb{E}[R] = -100000 + \frac{34000}{1.1} + \frac{34000}{1.1^2} + \frac{34000}{1.1^3} + \frac{34000}{1.1^4} + \frac{34000}{1.1^5} = 28886.75$$

2 W 8-2

2.1 part A

We now want to conduct a sensitivity analysis. Let the units sold be $u \in (70000, 100000, 130000)$. Build a table.

Year	0	High Sales	Medium Sales (base case)	Low Sales
Sales		130000	75000	52500
- Costs		−42500	−35000	−27500
- Depreciation		−20000	−20000	−20000
EBIT		67500	20000	5000
- taxes		−20250	−6000	−1500
Net Income		47250	14000	3500
+ Depreciation		20000	20000	20000
Operating CF		67250	34000	23500
- capital spending	−100000	0	0	0
- change in NWC		0	0	0
Project CF	−100000	67250	34000	23500
NPV		154930.41	28886.75	−10916.51

We could also back-solve these equations to find how many sales we need to break even. This comes out to 75485 sales per year.

2.2 part B

We now want to do a scenario analysis. The idea behind a scenario analysis is to look at several changes in economic conditions to see how your cashflows might change. (e.g. scenario 1 here has high demand, higher per unit price, and lower variable costs). [ran out of time in class for this problem]